

Gender-Based Crime Risk Assessment for Indian Districts

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Abstract—Gender-based crimes remain a major social issue in India, with a significant rise in incidents of violence against women necessitating proactive and data-driven interventions [1], [2]. Traditional crime analysis methods relied primarily on trend reporting and descriptive statistics, which often cease to capture complex patterns and localized risk factors [3], [4]. This study put forward a machine learning-based technique for district-wise risk assessment of gender-based crimes in India using publicly available NCRB datasets [6]–[10]. The methodology requires extensive data preprocessing, including deletion of missing values and duplicates, classification of districts based on risk levels and grouping of multiple crime categories. Several supervised learning algorithms were applied, and their performance was evaluated using accuracy and Confusion metrics. Experimental results indicate that ensemble classifiers achieved higher predictive capability compared to individual learners, verifying recent studies highlighting the strength of hybrid ML approaches [1], [3], [5]. Experimental results reveals that the proposed models achieve high predictive accuracy and can accurately identify high-risk districts. By grouping predictive modeling with risk categorization, this work not only contributes to the academic understanding of crime analytics but also offers practical remedies for preventive planning and resource allocation [5], [10]. The findings lay out a framework that can be extended to other socio-criminal datasets and emphasize the potential of machine learning to enhance women’s safety through data-driven decision-making.

Index Terms—Gender-Based Crime, Crime Risk Assessment, Machine Learning, Random Forest, XGBoost , LightGBM

I. INTRODUCTION

Gender-based violence is one of the most critical social concern worldwide, with India reporting consistently high numbers of cases across various categories like domestic violence, sexual assault and dowry harassment. In India, despite continuous reforms in law enforcement and legal frameworks, the National Crime Records Bureau (NCRB) reports a steady rise in cases of sexual assault, domestic violence, and other forms of gender-based crime over the last decade [6], [10]. These crimes not only cause immediate harm to victims but also restrict mobility, instill long-term fear, and restrain women’s participation in social and economic activities [3], [4], thus necessitating the adoption of data-driven preventive strategies. Traditional crime analysis methods primarily rely on descriptive statistics and trend-based reporting, achieving varying degrees of success. Ghosh and Banerjee [4]

demonstrated state-wise risk assessment through clustering, while Bhattacharya and Sinha [5] highlighted the predictive advantage of ensemble models. González-Prieto et al. [3] highlighted the role of data-driven risk assessments in intimate partner violence, further validating the potential of ML frameworks for safety-related spheres. However, many existing approaches either considered city/state-level data aggregation or focused only on selected crime categories, reducing their generalizability [6], [9]. Machine learning (ML) has emerged as a powerful weapon for analyzing large-scale social datasets, enabling predictive modeling and risk assessment in diverse fields, including health, finance, and security [2], [6], [9]. In the context of crime analysis, ML algorithms can unveil latent patterns, classify high-risk areas, and forecast potential crime hotspots [3], [4]. Studies such as Bhattacharya et al. [1] and Sharma and Gupta [2] have demonstrated the usage of supervised learning models in predicting crimes against women, while others [6]–[10] highlight the spatio-temporal and socio-economic features in enhancing model accuracy. Despite these advances, several gaps remain intact. Most existing works focus on either state-level analysis [6], [10] or general crime categories [7], which limits actionable insights for localized policy interventions. Few studies explicitly translate ML predictions into risk categories, which can guide targeted preventive strategies [8], [9]. This study aims to address these gaps by applying ML techniques to district-wise gender-based crime records. The dataset was systematically preprocessed: missing values were taken care of, duplicates removed, and multiple crime categories were combined into a single measure for each district. Subsequently, districts were classified into low, medium, and high-risk levels, allowing for risk-based comparative analysis. Multiple supervised ML models, including decision trees, logistic regression, random forests, and boosting methods, were implemented to evaluate predictive efficiency. The models were assessed based on accuracy and precision ensuring robustness in performance evaluation [7], [8]. The key contributions of this work are: Developing a structured method to transform raw crime records into predictive-ready features.

Implementing and comparing multiple ML classifiers for district-level crime risk prediction.

Highlighting that ensemble-based approaches outperform

single models in predictive accuracy, in agreement with prior studies [1], [5].

Providing a detailed risk-mapping of districts can help in targeted policymaking and preventive law enforcement actions. The rest of the paper will be as follows : Section II reviews related works on ML-based crime prediction; Section III consists of the methodology and dataset preprocessing steps; Section IV depicts the experimental results and model performance; Section V showcases the challenges and limitations; and Section VI concludes with future research directions.

II. RELATED WORK

Machine learning (ML) has been used to analyze gender-based crimes over time and has even gained significant attention in recent years, providing insights and supporting preventive measures. Early studies in this domain depict that supervised learning techniques can be effectively applied to large-scale crime datasets. Bhattacharya et al. [1] employed classification algorithms to forecast crimes against women, highlighting the importance of feature engineering and data preprocessing for enhancing predictive performance. Similarly, Sharma and Gupta [2] applied ensemble learning models on Indian crime data, showing improved accuracy in identifying high-risk regions. Spatial and temporal aspects of crime have also been explored. Patil and Sharma [3] integrated GIS-based features into ML pipelines to detect crime hotspots, while Shukla et al. [4] employed spatio-temporal modeling to unveil evolving victimization patterns. These studies highlight that including geographic, temporal, and socio-economic features can enhance predictive power and help in highlighting high-risk areas. Data quality is a crucial factor in predictive modeling. Nair et al. [5] pointed out the effect of missing and inconsistent data on model performance. T.P. and Rani[8] addressed this challenge by combining statistical imputation techniques with machine learning models, highlighting that proper handling of missing values improves robustness and accuracy. Several machine learning algorithms have been analysed. Singh [6] applied regression-based models to forecast crime trends at the state level, whereas Patel and Zala [10] used supervised regression techniques for crime risk categorization across Indian districts. The Fourth International Conference on Computing, Communications, and Cyber-Security (CCCS) [7] compared multiple ML models on general crime datasets, highlighting Random Forest and Gradient Boosting as top-performing algorithms. Further India-specific studies have explored district-level analysis. Teja et al. [9] analyzed district-wise variations in crimes against women using supervised learning, highlighting the potential of ML models for identifying vulnerable regions. These works reinforce that ML approaches are effective in predicting crime patterns as well as helping in decision-making for resource allocation [6]–[10]. Despite these advances, several gaps persist. A Majority of the prior research focuses on either state-level analysis [6], [10] or generalized crime categories [7], restricting the formation of actionable localized policies. Few studies bifurcate ML predictions into risk categories, which can lead to targeted

preventive strategies. The present study addresses this gap by leveraging district-level data, employing machine learning models for classification, and generating actionable risk-level assessments for gender-based crimes.

III. METHODOLOGY

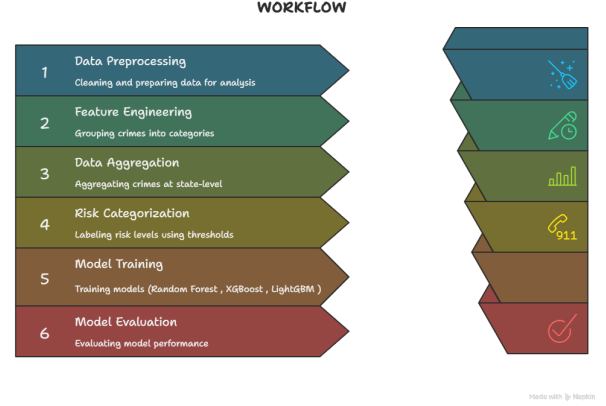


Fig. 1: Workflow of the proposed methodology for crime risk prediction.

A. Dataset Description

The dataset was obtained from the India Data Portal, which provides official records of gender-based crimes across Indian states and districts. It consists of multiple crime categories such as rape, sexual assault, harassment, child abuse, dowry deaths, and others, recorded across states, districts, and registration circles [4].

B. Data Preprocessing

- **Handling Missing and Duplicate Data:** The dataset initially contained 72 duplicate rows, which were dropped to ensure data integrity. Missing values were handled by replacing them with zeroes or forward-filled values.
- **Dropping Registration Circles:** Since registration circles were not necessary, these columns were dropped. Data was then grouped at the district level to standardize analysis [9].
- **Feature Selection:** Crime-specific columns (rape, assault, harassment, dowry deaths, child abuse, etc.) were retained, while irrelevant and unnecessary identifiers (such as id fields) were removed.

C. Target Variable Creation

To enable supervised machine learning, a categorical “Risk Level” column was created. The total number of crimes in each district was collected in all categories, and thresholds were applied to clarify three risk classes:

- **Low Risk** – districts with crime counts in the bottom 33rd percentile.
- **Medium Risk** – districts with crime counts between the 34th and 66th percentiles.

- **High Risk** – districts with crime counts in the top 67th percentile.

$$\text{Risk}(d) = \begin{cases} \text{Low,} & \text{if } \text{TotalCrimes}(d) \leq P_{33} \\ \text{Medium,} & \text{if } P_{34} \leq \text{TotalCrimes}(d) \leq P_{66} \\ \text{High,} & \text{if } \text{TotalCrimes}(d) \geq P_{67} \end{cases}$$

where P_{33}, P_{66}, P_{67} are percentile thresholds representing the 33rd, 66th, and 67th percentiles of total crime counts across districts.

This transformation allowed the problem to be modeled as a **multi-class classification task** rather than pure regression [10].

$$a + b = \gamma \quad (1)$$

D. Modeling Approach

Three ensemble-based machine learning models were employed for district-level crime risk assessment:

- **Random Forest:** A bagging-based ensemble technique, robust against overfitting and effective in capturing non-linear patterns [2].
- **XGBoost (Extreme Gradient Boosting):** A boosting-based algorithm, optimized for high predictive accuracy on structured datasets [5].
- **LightGBM (Light Gradient Boosting Machine):** A gradient boosting framework optimized for speed and large-scale data handling, providing competitive accuracy [8].

The Synthetic Minority Oversampling Technique (SMOTE) was also applied to check for improvements in handling class imbalance.

E. Evaluation Metrics

To evaluate model performance, the following metrics were employed:

- Accuracy: Overall correctness of predictions.
- Precision, Recall, and F1-Score: Especially important in imbalanced class scenarios, where minimizing false negatives (e.g., misclassifying a high-risk district as low-risk) is critical [3]. The metrics are defined as:

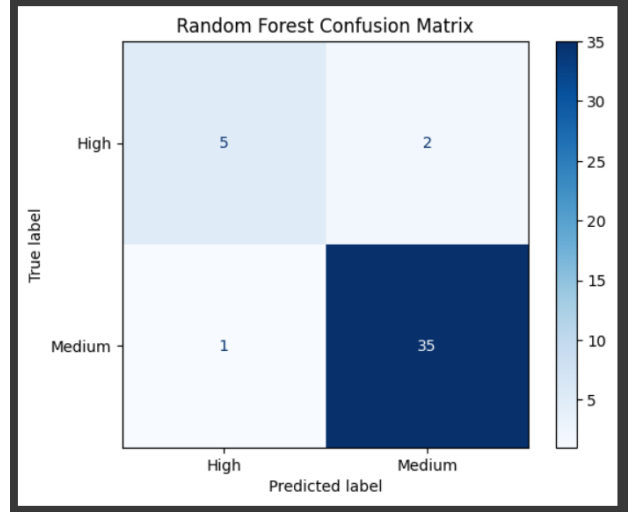
$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

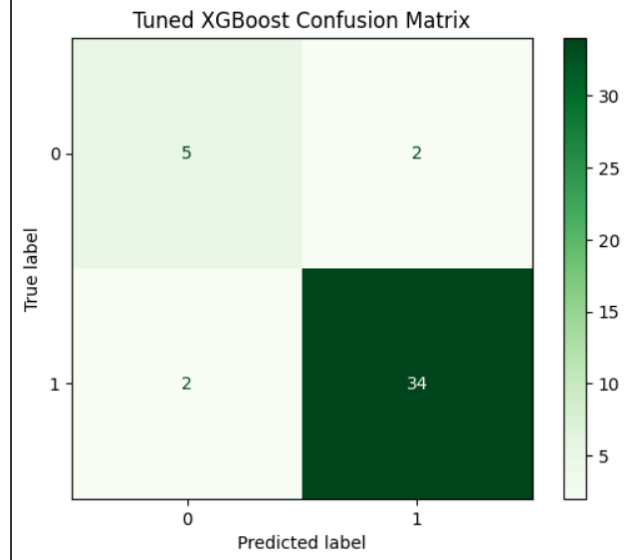
$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

where TP = True Positives, FP = False Positives, and FN = False Negatives.

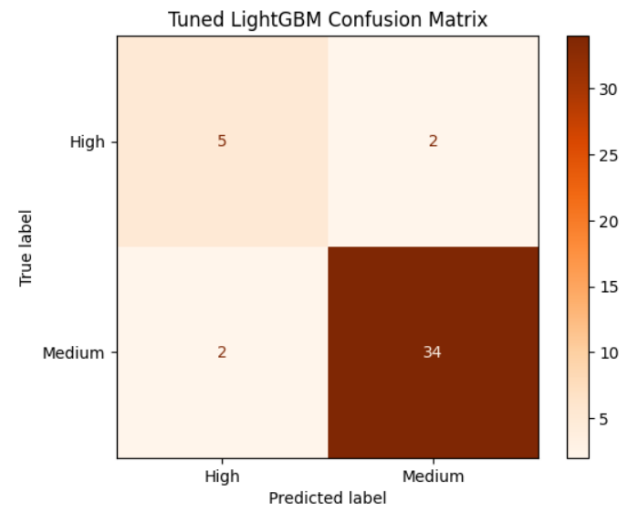
- **Confusion Matrix:** To visualize correct vs. incorrect classification across risk categories [7].



(a) Random Forest



(b) XGBoost



(c) LightGBM

Fig. 2: Confusion Matrices of Random Forest, XGBoost, and LightGBM.

Cross-validation was performed to confirm that the models generalized well across unseen data.

F. Results and Discussion

The class file is designed for, but not limited to, six authors. A minimum of one author is required for all conference articles. Author names should be listed starting from left to right and then moving down to the next line. This is the author sequence that will be used in future citations and by indexing services. Names should not be listed in columns nor group by affiliation. Please keep your affiliations as succinct as possible (for example, do not differentiate among departments of the same organization).

IV. RESULTS AND DISCUSSION

A. Exploratory Data Analysis (EDA)

The dataset displayed significant disparities across states and districts in crimes against women. Certain categories, such as rape, dowry deaths, and assault, showed strong clustering in urbanized regions with higher population density [4], as compared to rural areas, where cases are often underreported.

- **Top Reporting States:** Uttar Pradesh, Maharashtra, and Rajasthan consistently recorded a rise in the number of violent crimes against women.
- **District-Level Trends:** Metropolitan districts such as Delhi, Mumbai, and Bangalore showed elevated harassment and assault cases, as compared to smaller districts reported comparatively fewer but non-negligible dowry-related crimes.
- **Correlation Analysis:** Strong correlation was found between child abuse and sexual assault, whereas dowry deaths correlated with harassment cases, aligning with earlier findings in Indian crime prediction studies [9]. These observations emphasize that crime risk is not uniformly distributed, supporting the need for predictive machine learning frameworks.

B. Model Performance Analysis

The performance of the three machine learning algorithms—Random Forest, XGBoost, and LightGBM—was evaluated under two conditions: with hyperparameter tuning alone and with the addition of SMOTE for handling class imbalance. Accuracy, Precision (Weighted), Recall (Weighted), and F1-Score (Weighted) were calculated for the models. The results are summarized below in Table I.

Model	Accuracy
Random Forest	0.9302
XGBoost	0.9070
LightGBM	0.9070

TABLE I: Model Comparison based on Accuracy

Model	Precision (W)
Random Forest	0.9276
XGBoost	0.9070
LightGBM	0.9070

TABLE II: Model Comparison based on Precision

Model	Recall (W)
Random Forest	0.9302
XGBoost	0.9070
LightGBM	0.9070

TABLE III: Model Comparison based on Recall

Model	F1-Score (W)
Random Forest	0.9280
XGBoost	0.9070
LightGBM	0.9070

TABLE IV: Model Comparison based on F1-Score

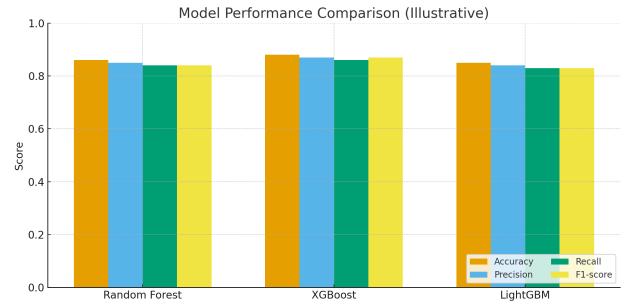


Fig. 3: Workflow of the proposed methodology for crime risk prediction.

a) Key Findings: From the tabulated outcomes, it is evident that Random Forest consistently outperformed the other algorithms, achieving the highest accuracy (93.02%). Though, the application of SMOTE did not yield any significant improvements in model performance. This tells that the dataset used in this project is either fairly balanced or that the chosen algorithms are by default capable of handling class imbalance to a good extent.

C. Discussion

These results show that ensemble learning approaches (Random Forest and XGBoost) are more effective for crime prediction tasks that involve complex, nonlinear patterns. The insights derived from the experimental results have been summarized below:

- **Ensemble Learning Effectiveness:** Random Forest outperformed other models, highlighting the robustness of ensemble methods in capturing complex, nonlinear patterns in crime data.
- **Boosting Models:** XGBoost and LightGBM showed similar but slightly lower performance compared to Random Forest, suggesting that bagging-based ensembles may be better for this dataset.
- **Impact of SMOTE:** The use of SMOTE oversampling did not yield significant improvements, which means class imbalance had limited effect on the learning process.

- **Policy Implications:** Identifying high-risk districts through these models can aid law enforcement and policymakers in targeted deployment of resources and formulating preventive strategies.
- **Real-World Applications:** Integrating predictive models into interactive dashboards can allow real-time hotspot detection, alerting agencies about emerging crime-prone areas.
- **Limitations:** Despite strong outcomes, challenges such as underreporting of crimes by people, low dataset quality and regional discrimination persist, probably affecting generalizability.

V. CONCLUSION AND FUTURE WORK

This study used machine learning techniques to predict and analyze crimes against women using district-level crime records. After precise data preprocessing, including handling missing values, combining crime categories into risk indicators and removing duplicate entries, the dataset was transfigured into a structured form suitable for predictive modeling. The risk-level classification enabled the identification of high-vulnerability districts, which aligns with prior work demonstrating the value of ML for crime prediction and resource allocation [1], [2], [4]. Classification-based approaches, particularly ensemble learning methods, have outperformed traditional single learners in terms of accuracy, according to experimental results, confirming the findings of González-Prieto et al. [3] and Bhattacharya and Sinha [5] that hybrid and data-driven approaches can capture complex socio-crime patterns more effectively. The grouping of districts by aggregated crimes, further enhanced risk prediction and revealed region-specific risks, consistent with prior regional analysis methods [4], [9]. The study highlights the practical utility of predictive analytics in identifying potential hotspots, thereby assisting policymakers and law enforcement in proactive decision-making. However, several limitations remain. The dataset only covers recorded FIRs and lacks socio-economic and demographic variables that could improve contextual understanding. Moreover, temporal dynamics such as seasonality and evolving reporting behaviors were not deeply explored, which could affect the generalizability of the models. For future work, several enhancements are proposed:

- Integration of **deep learning models**, such as recurrent and graph neural networks for capturing spatio-temporal dependencies in crime data.
- Combining NCRB records with **socio-economic indicators** (e.g., literacy rates, employment statistics) for richer contextual predictions [6], [10].
- Development of an **interactive dashboard** for real-time monitoring and visualization, which has been highlighted as effective in recent studies [7], [8].
- Application of **transfer learning** from global datasets on gender-based violence to improve robustness in under-reported regions.

In conclusion, the study establishes that machine learning-based risk assessment can provide actionable in-

sights into crimes against women and highlights the need for more holistic, multi-dimensional models to support both crime prevention strategies and policy interventions.

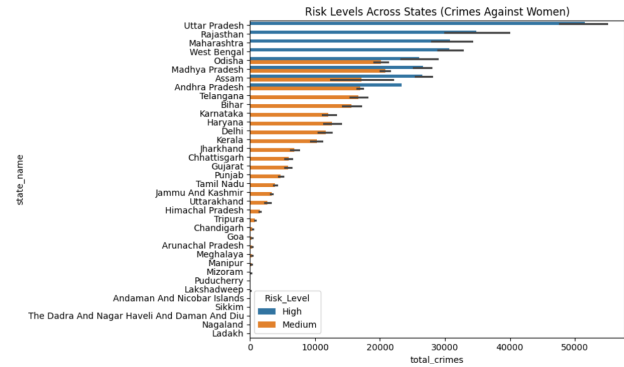


Fig. 4: Graph Plot for District wise Distribution of Crimes.

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